Multiplicity of zeroes of polynomials — Problem Set

Warm-up

1. State the multiplicity of the root $x = 3$ for the polynomial $P(x) = (x-3)^2(x+2)$.
2. If $P(x)$ has a root of multiplicity 2 at $x = \alpha$, what is the value of $P'(\alpha)$?
3. Find the zeros of $P(x) = x^2 - 6x + 9$ and state their multiplicity.
4. Given $P(x) = (x-1)^3(x+5)$, find $P'(1)$.
5. Determine if $x = 2$ is a multiple root of $P(x) = x^3 - 4x^2 + 4x$.
6. A polynomial $P(x)$ has a triple root at $x = -1$. List the three conditions involving $P(-1)$, $P'(-1)$, and $P''(-1)$.
7. Solve for k if $P(x) = x^2 + kx + 16$ has a double root.
8. Factorise $P(x) = x^3 - 3x^2 + 3x - 1$ and identify the multiplicity of its root.
9. If $P'(2) = 0$ and $P(2) = 0$, what is the minimum multiplicity of the root $x = 2$ in $P(x)$?
10. Find the derivative of $P(x) = x^4 - 4x^3$ and solve $P'(x) = 0$.

Skill-building

1. The polynomial $P(x) = x^3 - 3x + k$ has a repeated root. Find the possible values of k.
2. Given $P(x) = x^3 + ax^2 + bx + 8$ has a root of multiplicity 3 at $x = -2$. Find a and b.
3. Show that $P(x) = x^4 - 2x^3 + 2x - 1$ has a root of multiplicity 3 at $x = 1$.
4. The polynomial $P(x) = x^3 + px + q$ has a double root at $x = \alpha$. Show that $3\alpha^2 + p = 0$.
5. If $P(x) = x^n - 1$ has a root at $x = 1$, can this root ever have a multiplicity greater than 1? Justify using $P'(1)$.
6. Find the values of c and d such that $P(x) = x^4 + cx^2 + d$ has a triple root.
7. A quartic polynomial $P(x)$ has a triple root at $x = 2$ and $P(0) = -16$. Find the expression for $P(x)$ if it is monic.
8. The equation $x^3 - 9x^2 + 24x + k = 0$ has two equal roots. Find k.
9. Find the coordinates of the points on the curve $y = x^3 - 12x$ where the tangent is horizontal, and relate these to the multiple roots of $P(x) - k = 0$.
10. For $P(x) = x^4 - 4x + 3$, show that $x = 1$ is a double root and factorise $P(x)$ completely into linear factors.

Easier Exam Questions

1. (Cherrybrook Tech 2020 Q3) The polynomial $P(x) = x^3 - 12x^2 + 21x + k$ has a double root. What are the possible values of k?

   (A) $k = -1$ or $k = -7$

   (B) $k = 1$ or $k = 7$

   (C) $k = 10$ or $k = -98$

   (D) $k = -10$ or $k = 98$

2. (Ascham 2023 Q3) Which of the following polynomials $f(x)$ has a root of multiplicity 2 at $x = 1$ in the equation $f(x) = 0$?

   (A) $f(x) = x^2 - 5x + 4$

   (B) $f(x) = x^3 - 5x + 4$

   (C) $f(x) = x^4 - 5x + 4$

   (D) $f(x) = x^5 - 5x + 4$

3. (Abbotsleigh 2023 Q11a) The polynomial $P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$ has a double root at $x = 3$. **3**

   Find the value of a and b, where a and b are real numbers.

4. (Cheltenham Girls 2024 Q11f) Given that $P(x) = x^4 - 9x^3 + 25x^2 - 27x + 10$ has a double root, factorise $P(x)$ into linear factors. **2**

5. (Sydney Tech 2025 Q11d) Given that $P(x) = x^4 - 9x^3 + 25x^2 - 27x + 10$ has a double root, factorise $P(x)$ into linear factors. **2**

6. (St George Girls 2022 Q12c) The polynomial $P(x) = x^3 + ax^2 + bx - 12$ has a double root at $x = 2$. Find the values of a and b. **3**

Harder Exam Questions

1. (Sydney Girls 2025 Q5) It is given that $mx^4 + 3x^3 + 1 = 0$ has a double root. Which of the following is a correct expression for this double root?

   (A) $-\frac{9m}{4}$

   (B) $\frac{9m}{4}$

   (C) $-\frac{9}{4m}$

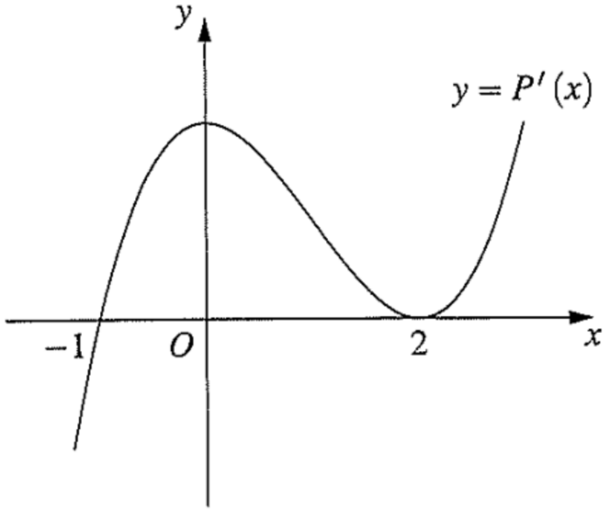
   (D) $\frac{9}{4m}$

2. (Fort St 2024 Q7)

   In the graph below, $y = P'(x)$ represents the first derivative of a polynomial $P(x)$ of degree 4, where $P(x)$ has a multiple root.

   Which of the following could be true about the polynomial $P(x)$?

   (A) $x = -1$ is a root of multiplicity 3.



(B) $x = 0$ is a root of multiplicity 2.

(C) $x = 2$ is a root of multiplicity 2.

(D) $x = 2$ is a root of multiplicity 3.

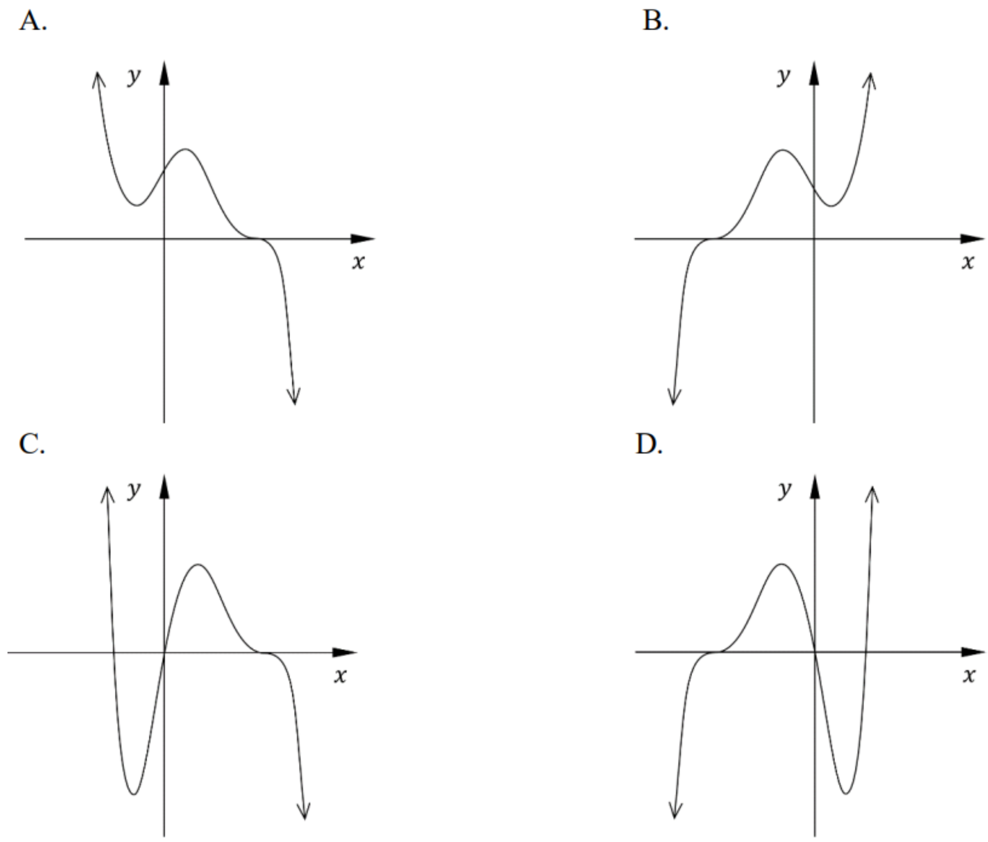
3. (Girraween 2025 Q11c) Consider the polynomial $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$. It is known that $P(x)$ has a triple root at $x = \alpha$.

   i) By using differentiation, find the value of α. **2**

   ii) Hence express $P(x)$ as the product of linear factors. **1**

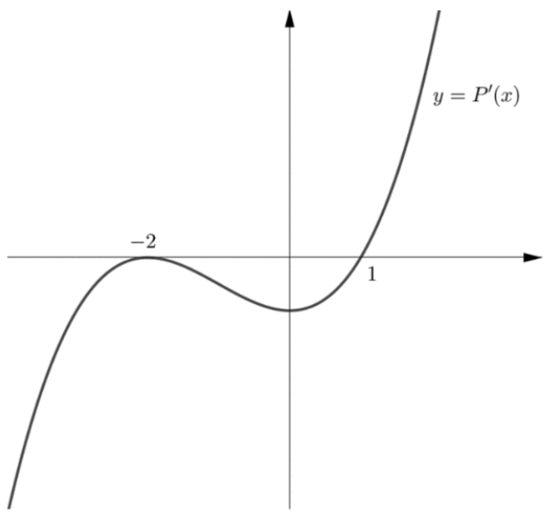
4. (Ascham 2024 Q12a) The polynomial $P(x) = x^3 + bx^2 + cx + 3$ has a root of multiplicity 2 at $x = -1$. **3**
   Find the values of b and c.

5. (Abbotsleigh 2024 Q3) A monic polynomial of degree 5 has one repeated root of multiplicity 3, and is divisible by $x^2 - x$. Which of the following could be the graph of $y = P(x)$?



6. (Girraween 2021 Q14b) Solve $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$, given that $P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$ has a triple zero. **3**

7. (Barker 2022 Q6) The graph of y = P'(x) is shown, where P(x) is a polynomial.
If x = -2 is a root of P(x), what is the multiplicity of this root for P(x)?



(A) 0
(B) 1
(C) 2
(D) 3